Def.

Let $G$ be a group and $A$ be a non-empty subset. A map

$$\alpha : G \times A \to A : (g, a) \mapsto g.a$$

is called a group action of $G$ on $A$ if it satisfies

1) For all $a \in A$, $1_G \cdot a = a$
2) For all $g_1, g_2 \in A$, and $a \in A$, $(g_1 g_2) \cdot a = g_1 \cdot (g_2 \cdot a)$

Prop. Given a group action $\alpha$ of $G$ on $A$, there exists a homomorphism

$$\varphi_\alpha : G \to S_A : g \mapsto \sigma_g \quad \text{where} \quad \sigma_g : A \to A : a \mapsto ga.$$

which is called the permutation representation associated to the given action.
Conversely, given a homomorphism $\varphi : G \to S_A$, there exists a group action

$$\alpha_\varphi : G \times A \to A : (g, a) \mapsto \varphi(g)(a)$$

Proof. We only need to check that the given map is well-defined,
Let $g \in G$ be fixed. Then, $\sigma_g \circ \sigma_{g^{-1}}(a) = \sigma_{g^{-1}} \circ \sigma_g(a) = a$, therefore $\sigma_g \in S_A$.
The converse is obtained by the definition of group action.

Note: The concept of group action is; for a set $A$, determind a permutation

$$A \to A ; a \mapsto g \cdot a$$

from each element $g \in G$, compatiable with the group structure of $G$,
in the sence ; the afforded homomorphism $\varphi_\alpha : G \to S_A$ preserves the operation.

**Def.** Let $G$ acts on $A$. For each $a \in A$, a subset
$$\mathrm{Orb}_G(a) \overset{\mathrm{def}}{=} \{g \cdot a \mid g \in G\} \subseteq A$$
is called the Orbit of $a$.

**Prop.** Let a group $G$ acts on a set $A$. Given $a \in A$, a map
$$G/G_a \to \mathrm{Orb}_G(a) : gG_a \mapsto g \cdot a$$
is bijection. In particular, if $G$ is finite, then
$$|G : G_a| = \#\mathrm{Orb}_G(a).$$
Proof. Given $g_1, g_2 \in G$,
$$g_1 G_a = g_2 G_a \Leftrightarrow g_1^{-1} g_2 \in G_a \Leftrightarrow g_1^{-1} g_2 a = a \Leftrightarrow g_1 a = g_2 a$$
So it is well-defined and injective. And, the surjectiveness is clear. $\blacksquare$

**Prop.** (Dummit, 4.1.9). Let $G$ acts transitively on the finite set $A$, and $H \trianglelefteq G$ be normal. Let $\mathcal{O}_1, \cdots, \mathcal{O}_r$ be the distinct orbits of $H$ on $A$

a). For each $g \in G$ and $i \in \{1, \cdots, r\}$, there is a $j$ such that $g\mathcal{O}_i = \mathcal{O}_j$. Further, $G$ is transitive on $\{\mathcal{O}_1, \cdots, \mathcal{O}_r\}$, therefore all $\mathcal{O}_i$ have same cardinality.

b). If $a \in \mathcal{O}_1$, then $|\mathcal{O}_1| = |H : H \cap G_a|$ and therefore $r = |G : HG_a|$.

Proof. Let $x \in \mathcal{O}_i$ Then, we can write $\mathcal{O}_i = \{hx_i \mid h \in H\}$. Given $g \in G$,
$$g\mathcal{O}_i = \{ghx_i \mid h \in H\} = \{h' g x_i \mid h' = ghg^{-1} \in H\} = \mathrm{Orb}_H(gx_i) = \mathcal{O}_j, \text{ for some } j.$$
Let $\mathcal{O}_1$ be fixed. Then, for any $\mathcal{O}_j$ with representative $x_j$, there is $g \in G$ such that $gx_1 = x_j$, so $g\mathcal{O}_1 = \mathcal{O}_j$, it proves a).

By the second-isomorphism theorem, given $a \in \mathcal{O}_1$,
$$HG_a/H \cong G_a/H \cap G_a$$
$$\begin{tikzcd} & G \arrow[d, dash] & \\ & HG_a \arrow[dl, dash] \arrow[dr, dash] & \\ H \arrow[dr, dash] & & G_a \arrow[dl, dash] \\ & H \cap G_a & \end{tikzcd}$$
So, $|\mathcal{O}_1| = |H : H \cap G_a|$ and $|A| = r \cdot |\mathcal{O}_1|$ implies
$$|A| = |G : G_a| = |G : HG_a||HG_a : G_a| = |G : HG_a| \cdot |H : H \cap G_a| = |G : HG_a| \cdot |\mathcal{O}_1|.$$
gives the result.

「Thm.

Let $G$ be a group and $H \leq G$ be a subgroup.
Consider $G$ acts on $\{gH \mid g \in G\}$ by left multiplication, and affored homomomorphism

$$\pi_H : G \to S_{G/H} : g \mapsto \sigma_g \text{ where } \sigma_g(xH) = gxH.$$

Then $G$ acts transitively on $A$ and the stabilizer $G_{1 \cdot H} = H$, and

$$\ker \pi_H = \bigcap_{x \in G} xHx^{-1} \leq H$$

is the largest normal subgroup of $G$ contained in $H$.

Proof. Given $aH, bH \in A$, $bH = ba^{-1}(aH)$, so $aH$ and $bH$ lie in same orbit.
And, since $gH = H \iff g \in H$,

$$G_{1 \cdot H} = \{g \in G \mid gH = H\} = H.$$

To show the last assertion, by definition,

$$\ker \pi_H = \bigcap_{x \in G} G_{xH}$$

and $G_{xH} = \{g \in G \mid gxH = xH\} = \{g \in G \mid x^{-1}gx \in H\} = xHx^{-1}$.
Moreover, $\ker \pi_H \trianglelefteq G$ because it is kernel of the homomomorphism,
and for any normal subgroup $N \trianglelefteq G$ of $G$ contained in $H$,

for all $x \in G$, $N = xNx^{-1} \leq xHx^{-1}$

therefore $N \leq \ker \pi_H$. ☑

「Prop. If $H \leq G$ has finite index $n = |G:H|$, then there is a normal subgroup $K \trianglelefteq G$ s.t

$$K \leq H \text{ and } |G:K| \,\big|\, n!$$

Proof. The kernel of $\pi_H : G \to S_{G/H} : g \mapsto (xH \mapsto gxH)$ is contained in $H$. And $G/\ker \pi_H \cong \mathrm{Im}\, \pi_H \leq S_{G/H} \cong S_n$. Thus $|G:\ker \pi_H| \mid n!$, and

$$|G : \ker \pi_H| = |G:H| \cdot |H : \ker \pi_H| = n \cdot |H : \ker \pi_H|$$

So $|H : \ker \pi_H|$ divides $(n-1)!$.

Thm.
Let $G$ be a finite group of order $n$, and let $p$ be the smallest prime dividing $n$.
Then, every subgroup of index $p$ is normal.
Proof. Let $H \leq G$ with $|G:H| = p$. Let $G$ acts on left cosets of $H$ and

$$\pi_H : G \to S_{G/H}$$

be the afforded permutation of the given action.
Then, $\operatorname{Ker} \pi_H \leq H$, let $K = \operatorname{Ker} \pi_H$ and $k = |H:K|$. Then,

$$|G:K| = |G:H| \cdot |H:K| = pk.$$

Since $G/K \cong \operatorname{Im} \pi_H \leq S_p$, so $pk$ divides $p!$ by the Lagrange's Theorem.
Therefore, $k$ divides $(p-1)!$. But all prime factors of $(p-1)!$ are less than $p$,
by the minimality of $p$, $k$ must be $1$, that is, $H = K = \operatorname{Ker} \pi_H \trianglelefteq G$ ∎

Note: let $G$ acts on finite set $A$ which has $m$ elements.
Then, the afforded permutation $\varphi : G \to S_m$ satisfies:

$$G/\operatorname{Ker} \varphi \cong \operatorname{Im} \varphi \leq S_m \text{ implies } |\operatorname{Im} \varphi| \,\big|\, |G| \text{ and } |\operatorname{Im} \varphi| \,\big|\, m!$$

So, if $p$ is the smallest prime factor of $|G|$, then

$$m < p \Rightarrow \operatorname{Im} \varphi = 1 \quad \text{and} \quad m = p \Rightarrow |\operatorname{Im} \varphi| \in \{1, p\}.$$

In the case of the above theorem, the action is transitive, therefore $\operatorname{Im} \pi_H \neq 1$.
Therefore, $|\operatorname{Ker} \pi_H| = |H|$ and $\operatorname{Ker} \pi_H \leq H$, so proved.

Def. Let $G$ be a group, and let $H \leq G$ be a subgroup.
A subgroup $\operatorname{Core}_G(H) \overset{\text{def}}{=} \bigcap_{g \in G} gHg^{-1}$ is called the core of $H$ in $G$.
(It is normal).
Note: the theorem above gives

$$H \trianglelefteq G \iff H = \operatorname{Core}_G(H).$$

「Problems in dummit, 4.2.

「Thm. If $G$ is a group of order $p^n$ where $p$ is prime and $n \in \mathbb{N}$, then every subgroup of index $p$ is normal.

Proof. Let $H \leq G$ be the subgroup with $|G:H| = p$.
Then, the Homomorphism afforded by the left coset action

$$\varphi : G \to S_{G/H} ; g \mapsto (xH \mapsto gxH)$$

Satisfies

$$G/\ker \varphi \cong \operatorname{Im} \varphi \leq S_{G/H}$$

So, $|G/\ker \varphi|$ divides $p!$, but since $p$ is prime, $|G/\ker \varphi| \in \{1, p\}$.
Since $\ker \varphi \leq H \neq G$, $\ker \varphi \neq G$, so $|G/\ker \varphi| = p$, that is, $|\ker \varphi| = p^{n-1}$.
Consequently, $H = \ker \varphi \trianglelefteq G$ is normal. ☑

Note: If $G$ is group of order $p^2$, then it has order $p$ element $x \in G$ by Cauchy's theorem and $\langle x \rangle \trianglelefteq G$.

「Thm. Group of order 6 are only $\mathbb{Z}/6\mathbb{Z}$ and $S_3$.

Proof. If $G$ is abelian, then the Cauchy theorem gives order 2 element $x$ and order 3 element $y$. Since $x$ and $y$ commute, $|xy| = 6$, so $G = \langle xy \rangle \cong C_6$.
Assume that $G$ is non-abelian. By the Cauchy theorem again, there exists order 2 element $x$. Let $H = \langle x \rangle$. Then, the homomorphism

$$\pi_H : G \to S_{G/H} : g \mapsto (xH \mapsto gxH)$$

Satisfies that $G/\ker \pi_H \cong \operatorname{Im} \pi_H \leq S_{G/H} \cong S_3$.
If $H$ is normal, then for all $g \in G$, $gxg^{-1} \in \langle x \rangle$, that is, $gxg^{-1} = x$, so $x$ commutes with order 3 element $y$, it implies $\langle xy \rangle = 6$, it contradicts.
Therefore $H$ is non-normal, so $\ker \pi_H \lneq H$, the trivial subgroup. Now,

$$G/\ker \pi_H \cong G \cong \operatorname{Im} \pi_H \leq S_{G/H} \cong S_3$$

and since $G$ and $S_3$ both are order 6, thus $G \cong S_3$ ☑

Thm. Let $G$ be a finite group and let $\pi : G \to S_G$ be the left regular representation. Given $x \in G$, the permutation $\pi(x)$ is a product of $|G|/|x|$ numbers $|x|$-cycles.

「Conjugation action.

「Def.

Let $G$ acts $G$ itself by Conjugation:

$$G \times G \to G : (g, x) \mapsto gxg^{-1}.$$

With an affored Homomorphism

$$\pi : G \to S_G : g \mapsto (x \mapsto gxg^{-1})$$

Observe that:

$$\ker \pi = \{g \in G \mid gxg^{-1} = x \text{ for all } x \in G\} = Z(G)$$

$$\operatorname{Im} \pi = \operatorname{Inn}(G) \leq \operatorname{Aut}(G) \leq S_G.$$

By the first-isomorphism theorem,

$$G/Z(G) = G/\ker \pi \cong \operatorname{Im} \pi = \operatorname{Inn}(G)$$

And, the stabilizer of $x \in G$

$$G_x = \{g \in G \mid gxg = x\} = C_G(\{x\}).$$

and the Orbit

$$\operatorname{Orb}_G(x) = \{gxg^{-1} \mid g \in G\}.$$

「Prop. Let $G$ be a group. Then,

$$|G| = |Z(G)| + \sum_{i=1}^{r} |G : C_G(x_i)|$$

Where $x_1, \cdots, x_r$ are reperesentatives of Conjugacy classes not Contained in the center.

「Conjugate on subgroups.

Let $G$ acts on $\{H \leq G\}$ by conjugation

$$G \times \{H \leq G\} \to \{H \leq G\} : (g, H) \mapsto gHg^{-1}.$$

With the afforded Homomorphism

$$\pi_S : G \to S_{\{H \leq G\}} : g \mapsto (H \mapsto gHg^{-1})$$

Obeserve that: $Stab(H) = \{g \in G \mid gHg^{-1} = H\} = N_G(H)$.

Directly, $\#Orb_G(H) = \#\{gHg^{-1} \mid g \in G\} = |G : N_G(H)|$.

In Particular, when $G$ is finite group of order $p^n \cdot m$, there is order $p^n$ subgroup $P \leq G$,

$$Orb_G(P) = Syl_p(G) = \{P, g_2 P g_2^{-1}, \cdots, g_{n_p} P g_{n_p}^{-1}\}.$$

「Conjugate on a normal subgroup.

Let $G$ acts on a normal subgroup $H \trianglelefteq G$ by conjugation

$$G \times H \to H : (g, h) \mapsto ghg^{-1}$$ (It is well-defined by the normality.

With the afforded Homomorphism

$$\pi : G \to S_H : g \mapsto (h \mapsto ghg^{-1}).$$

Observe that: $\ker \pi = C_G(H)$ and $Inn(H) \leq Im\, \pi \leq Aut(H)$.

「Prop. Given a subgroup $H \leq G$,

$$N_G(H)/C_G(H) \cong Im\, \pi \leq Aut(H).$$

In particular, $G/Z(G) \hookrightarrow Aut(G)$.

「Summary.

Let $G$ be a group, and $H < G$ be a subgroup, $N \trianglelefteq G$ be a normal subgroup.

「Left Coset Action. $G$ acts on $G/H$ by $(g, xH) \mapsto gxH$.

Afforded Homomorphism: $\pi_H : G \to S_{G/H} : g \mapsto (xH \mapsto gxH)$.

Kernel: $\operatorname{Ker} \pi_H = \operatorname{Core}_G(H) = \bigcap_{x \in G} xHx^{-1}$

Stabilizer: $\operatorname{Stab}(xH) = xHx^{-1}$.

Orbit: $\operatorname{Orb}(H) = G/H$, transitive.

「Conjugate itself. $G$ acts on $G$ by $(g, h) \mapsto ghg^{-1}$.

Afforded Homomorphism: $\varphi : G \to S_G : g \mapsto (h \mapsto ghg^{-1})$

Kernel: $\operatorname{Ker} \varphi = Z(G)$

Image: $\operatorname{Im} \varphi = \operatorname{Inn}(G) \leq \operatorname{Aut}(G) \leq S_G$

Stabilizer: $\operatorname{Stab}(x) = C_G(x)$

Orbit: $\operatorname{Orb}(x) = Cl(x) = \{gxg^{-1} \mid g \in G\}$.

「Conjugate on Subgroups. $G$ acts on $\{K \leq G\}$ by $(g, K) \mapsto gKg^{-1}$.

Afforded Homomorphism: $\psi : G \to S_{\{K \leq G\}} : g \mapsto (K \mapsto gKg^{-1})$.

Kernel:

Image:

Stabilizer: $\operatorname{Stab}(K) = N_G(K)$

orbit: $\operatorname{Orb}(K) = \{gKg^{-1} \mid g \in G\}$

Prop. (Dummit. Sec 4.3.(a)).
Let $G$ be a group and $H \trianglelefteq G$ be a normal subgroup.
Let $K$ be a conjugacy classes of $G$ contained in $H$, and let $x \in K$.
Then, $K$ is a union of $|G:HC_G(x)|$-conjugacy classes of equal size in $H$. That is,

$$Cl_G(x) = \bigcup_{i=1}^{k} Cl_H(x_i) \quad \text{where } k = |G:HC_G(x)|,\ |Cl_H(x_i)| = \frac{|Cl_G(x)|}{|G:HC_G(x)|}.$$

Proof. Notice that since $H$ is normal, for any $x \in H$, $Cl_G(x) \subseteq H$.
Let $G$ acts $G$ itself by conjugation and consider the afforded homomorphism

$$\pi : G \to S_G : g \mapsto (h \mapsto ghg^{-1})$$

Then, $\operatorname{Ker}\pi = \{g \in G : ghg^{-1} = h \ \forall h \in G\} = Z(G)$, and $Orb_G(x) = Cl_G(x)$, and
$Stab_G(x) = C_G(x)$. Therefore, $|G:C_G(x)| = |Cl_G(x)|$.
Meanwhile, similarly, consider the afforded homomorphis

$$\pi_H : H \to S_H : h \mapsto (k \mapsto hkh^{-1}).$$

Since $C_H(x) = H \cap C_G(x)$ by just definition,

$$|HC_G(x)| = \frac{|H|\cdot|C_G(x)|}{|H \cap C_G(x)|} = \frac{|H|\cdot|C_G(x)|}{|C_H(x)|}$$

Application to finite group analysis.

Prop. Every group of order of $pq$ where $p$ and $q$ are distinct prime with $p \nmid q-1$ is abelian